

Operating Systems and Networking

Network basics

IDENTIFICATION

CODE : IF-3-RE
ECTS : 2.0

HOURS

Lectures : 10.5 h
Seminars : 8.0 h
Laboratory : 8.0 h
Project : 0.0 h
Teacher-student
contact : 26.5 h
Personal work : 25.0 h
Total : 51.5 h

ASSESSMENT METHOD

Personal works associated to each TD and lab and a final exam [documents are allowed]

TEACHING AIDS

Paper copy of the slides of the course. Basic exercises and Lab description. Videos and on line exercises associated to key points / practical works will be available on the Moodle Platform [in french].

TEACHING LANGUAGE

French

CONTACT

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AIMS

This course aims at providing the necessary knowledge and competencies to design and manage a distributed infrastructure, supporting corporates information system (including industrial and SCADA computing resources). These technical knowledge and competencies include:

- (1) understanding protocol engineering models and methods
 - (2) providing the necessary knowledge on telecommunication infrastructure (from signal processing techniques and constraints to the introduction routing principles)
 - (3) understanding and being able to implement a method to design and configure a distributed infrastructure
- TDs and labs provide realistic use-cases to support a learning by example strategy.

CONTENT

Part 1: Protocol engineering

- 1.1: Models used to implement communication stacks including finite state automata based descriptions, activity encapsulation and data structure (i.e. PDU, SDU and PCI) organisation
- 1.2: Specification methods including automata meta-model, SDL descriptions, organisation of a testing strategy
- 1.3: Algorithmic patterns including variables, windows and time management

Part 2: Telecommunication systems and infrastructure

- 2.1: Information transmission and signal processing including signal processing basics, modulation techniques and how to provide a functional design of a modem and of an interface protocol
- 2.2: LAN organisation and Medium Access Control methods
- 2.3: Routing principles and introduction to IP

Part 3: Method to design and configure a distributed infrastructure

- 3.1: Requirements and traffic analysis
- 3.2: Wired infrastructure organisation and interconnection principles
- 3.3: Configuration management & Introduction to DHCP and DNS protocols

TD and labs use a realistic use case (namely a network infrastructure designed for a SME) to support practical works. Simulators and virtual machines are used to allow practicing network design and configuration as well as introducing network management basic toolset.

BIBLIOGRAPHY

- [1] G. Pujolles. Les Réseaux
- [2] A. Tannenbaum. Réseaux : architecture, protocoles, applications.
- [3] G. Beuchot. Téléinformatique : Tome 1. Polycopié INSA.
- [4] L. Toutain. Réseaux locaux et Internet

PRE-REQUISITE

3IF-RE1 - Network programming skills.

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